

NIRVANA – Master Specification

Volume 5: Recommendation & Optimization Engine

Objective

The Recommendation Engine is the deterministic decision-making core of Nirvana. AI never performs calculations; it explains results generated by the engine.

Decision Pipeline

User Profile → Financial Data → Risk Engine → Behaviour Engine → Goal Engine → Asset Allocation → Tax Optimization → Rebalancing → Withdrawal Logic → AI Explanation.

Inputs

Age, income, expenses, cash flow, dependents, goals, liabilities, assets, tax regime, insurance, credit score, investment horizon, risk score, behavioural biases, market conditions and macroeconomic indicators.

Risk Engine

Assign a quantitative risk score (0-100) using questionnaire responses, investment horizon, income stability, liquidity needs and behavioural indicators. Output Conservative, Moderate, Aggressive or Very Aggressive.

Behavioural Finance

Detect biases such as loss aversion, anchoring, herd behaviour, recency bias, overconfidence and mental accounting. Adjust recommendations by including behavioural coaching rather than changing deterministic calculations.

Goal Engine

For each goal compute inflation-adjusted target value, funding gap, required SIP, expected corpus, probability of success and priority ranking.

Asset Allocation

Recommend strategic and tactical allocation across equity, debt, gold, cash, alternatives and government schemes. Allocation should respect user risk profile, liquidity needs and concentration limits.

Optimization

Use deterministic optimization to maximize expected goal success while minimizing unnecessary tax, excessive concentration, liquidity risk and portfolio drift. Support Monte Carlo simulation for probability estimates.

Portfolio Rebalancing

Continuously compare current allocation against target allocation. Recommend partial rebalancing only when thresholds are breached or goals materially change.

Recommendation Ranking

Rank every recommendation by estimated financial impact, urgency, implementation complexity and confidence. Present only the highest-value actions first.

Human vs AI

Maintain a comparison layer showing advisor recommendation, AI explanation, final decision and rationale. Human overrides should be recorded for auditability.

Output

Generate a structured action plan with investment recommendations, insurance gaps, tax opportunities, debt actions, withdrawal advice, expected outcomes and supporting explanations.